

Final Exam - Solution outlines

1. For each of the following problems, we suppose that $X_1, X_2, \dots$ are independent and identically distributed (i.i.d.) random variables with mean 0 and variance 1.

- (a) (5 points) What is the expected value of the random variable $Y = X_1 + 2X_2 + 3X_3X_4$?

Solution: Using the linearity of expectation we get

$$E[X_1 + 2X_2 + 3X_3X_4] = E[X_1] + 2E[X_2] + 3E[X_3X_4]$$

Since X_3 and X_4 are independent, we also have $E[X_3X_4] = E[X_3]E[X_4]$. Since the random variables are mean zero, we get

$$E[X_1 + 2X_2 + 3X_3X_4] = E[X_1] + 2E[X_2] + 3E[X_3]E[X_4] = 0$$

- (b) (5 points) Find $\text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3 + 1)$.

Solution: The covariance of two random variables does not change if we add a constant to one of them, hence

$$\text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3 + 1) = \text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3).$$

Using $\text{Cov}(Y_1 + Y_2, Z) = \text{Cov}(Y_1, Z) + \text{Cov}(Y_2, Z)$ and $\text{Cov}(Y, Z) = \text{Cov}(Z, Y)$ repeatedly we get

$$\text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3) = \text{Cov}(X_1, 3X_2) + \text{Cov}(X_1, 4X_3) + \text{Cov}(2X_2, 3X_2) + \text{Cov}(2X_2, 4X_3)$$

Using $\text{Cov}(aY, Z) = a \text{Cov}(Y, Z)$ leads to

$$\text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3) = 3 \text{Cov}(X_1, X_2) + 4 \text{Cov}(X_1, X_3) + 6 \text{Cov}(X_2, X_2) + 8 \text{Cov}(X_2, X_3).$$

Independent random variables have covariance 0, and since X_1, X_2, X_3, X_4 are independent, we get

$$\text{Cov}(X_1 + 2X_2, 3X_2 + 4X_3) = 6 \text{Cov}(X_2, X_2) = 6 \text{Var}(X_2) = 6.$$

In the last step we used that the covariance of a random variable with itself is the variance.

- (c) (5 points) Suppose that we know that the moment generating function of X_1 is given by $M(t) = \frac{1}{1-t^2/2}$ whenever $|t| < \sqrt{2}$. What is the moment generating function of the random variable $Z = X_1 + X_2$ at $t = 1$?

Solution: Since X_1 and X_2 are independent, the MGF of their sum is just the product of the individual MGFs:

$$M_{X_1+X_2}(t) = M_{X_1}(t)M_{X_2}(t).$$

Since X_1 and X_2 have identical distributions, their MGFs are the same (and equal to $M(t)$):

$$M_{X_1+X_2}(t) = M_{X_1}(t)M_{X_2}(t) = M(t)^2$$

Plugging in $t = 1$ gives

$$M_{X_1+X_2}(t) = M(t)^2 = \frac{1}{(1-1/2)^2} = 4.$$

- (d) (5 points) Suppose that X_1 has standard normal distribution. Find $P(W > 0)$ for $W = X_1 + 2X_2 - X_3$.

Solution: We have seen in class that a linear combination of independent normals is also a normal. Hence W has normal distribution, the mean is equal to $E[X_1 + 2X_2 - X_3] = E[X_1] + 2E[X_2] - E[X_3] = 0$, and the variance is

$$\text{Var}(X_1 + 2X_2 - X_3) = \text{Var}(X_1) + 2^2 \text{Var}(X_2) + (-1)^2 \text{Var}(X_3) = 6.$$

(We used in the last step that the variance of the sum of independent random variables is the sum of the variances.) Hence

$$P(W > 0) = P(Z > 0)$$

where $Z \sim N(0, 6)$. Then $Z/\sqrt{6}$ is a standard normal and we get

$$P(Z > 0) = P\left(Z/\sqrt{6} > 0\right) = 1 - \Phi(0) = 1/2.$$

2. Jrue is a professional basketball player, averaging 16 points per game. Let X denote the number of points he scores in his next basketball game.

Note. You will estimate and/or approximate certain probabilities in some of the following problems. Justify the approach you use!

- (a) (5 points) What is the best upper bound you can provide for $P(X \geq 30)$?

Solution: Since X is a non-negative random variable, we can apply Markov's inequality:

$$P(X \geq 30) \leq \frac{E[X]}{30} = \frac{16}{30} = \frac{8}{15}.$$

- (b) (5 points) Suppose that we also learn that the variance of X is approximately 36. How can you improve on your upper bound on $P(X \geq 30)$?

Solution: We can use Chebyshev's inequality to estimate this probability. For this we first rewrite the event as an inequality on the deviation from the mean:

$$\begin{aligned} P(X \geq 30) &= P(X - E[X] \geq 30 - E[X]) = P(X - E[X] \geq 14) \leq P(|X - E[X]| \geq 14) \\ &\leq \frac{\text{Var}(X)}{14^2} = \frac{36}{14^2} = \frac{9}{49}. \end{aligned}$$

- (c) (5 points) Approximate the probability that in the next season (100 games, counting playoffs), Jrue will score more than 1700 points in total.

Note. You may assume that the number of points scored by Jrue in different games are independent and identically distributed.

Solution: Let $X_1, \dots, X_{100}$ denote the points scored in the various games next season, and let S denote their sum. Since $X_1, \dots, X_{100}$ are iid by assumption we may use the Central Limit Theorem to approximate the standardized version of S with a standard normal random variable Z . Using the fact that the random variables are iid we get

$$E[S] = 100E[X_1] = 1600, \quad \text{Var}(S) = 100 \text{Var}(X_1) = 3600 = 60^2,$$

so the standardized version of S is $\frac{S - E[S]}{\sqrt{\text{Var}(S)}} = \frac{S - 1600}{60}$. This leads to

$$P(S > 1700) = P\left(\frac{S - 1600}{60} > \frac{1700 - 1600}{60}\right) \approx P\left(Z > \frac{5}{3}\right) = 1 - \Phi(5/3).$$

3. We flip a fair coin. If it is heads, we choose a number uniformly at random from $[0, 1]$. If it is tails, we choose a number uniformly at random from $[0, 2]$. Let X denote the chosen number.

(a) (5 points) Find the probability that $X \leq 1/2$.

Solution: We will condition on the outcome of the coin flip. Let B be the event that the coin lands heads, and let $A = \{X \leq 1/2\}$. Then by the law of total probability we have

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c).$$

We have $P(B) = P(B^c) = 1/2$. From the setup we know that

$$P(A|B) = P(X < 1/2|B) = \frac{1/2}{1} = 1/2, \quad P(A|B^c) = P(X < 1/2|B^c) = \frac{1/2}{2} = 1/4.$$

This gives

$$P(A) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} = \frac{3}{8}.$$

(b) (5 points) Find the probability that we rolled heads given that $X \leq 1/2$.

Solution: Using the notation of part (a) we need $P(B|A)$, which we can compute using Bayes formula and the probabilities from part (a):

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)} = \frac{1/2 \cdot 1/2}{3/8} = \frac{2}{3}.$$

(c) (5 points) Find the cumulative distribution function of X . (Make sure that the function you provide is defined on the whole line.)

Solution: We need to compute $F(t) = P(X \leq t)$ for all values of t . Note that X can only take values from $[0, 2]$, hence $F(t) = 0$ for $t < 0$ and $F(t) = 1$ for $t \geq 2$.

For $t \in [0, 2)$ we can compute $P(X \leq t)$ by conditioning, just as in part (a).

$$P(X \leq t) = P(X \leq t|B)P(B) + P(X \leq t|B^c)P(B^c)$$

If we flip heads the conditional distribution of X is uniform on $[0, 1]$, hence $P(X \leq t|B)$ is equal to t for $t \in [0, 1]$ and equal to 1 if $t > 1$.

If we flip tails the conditional distribution of X is uniform on $[0, 2]$, hence for $t \in [0, 2)$ we have $P(X \leq t|B^c) = \frac{t}{2}$.

Hence for $t \in [0, 1]$ we have

$$F(t) = t \cdot 1/2 + \frac{t}{2} \cdot 1/2 = \frac{3}{4}t,$$

and for $t \in [1, 2)$ we have

$$F(t) = 1 \cdot 1/2 + \frac{t}{2} \cdot 1/2 = \frac{t+2}{4}.$$

This leads to

$$F(t) = \begin{cases} 0, & t < 0 \\ \frac{3}{4}t, & 0 \leq t \leq 1 \\ \frac{t+2}{4}, & 1 < t \leq 2 \\ 1, & 2 \leq t. \end{cases}$$

- (d) (5 points) Show that X is continuous, and find its probability density function.

Solution: The CDF of X is continuous everywhere, and it is differentiable apart from finitely many values. Hence X has continuous distribution, and the PDF can be computed by differentiating F :

$$f_X(t) = \begin{cases} \frac{3}{4}, & 0 \leq t \leq 1 \\ \frac{1}{4}, & 1 < t \leq 2 \\ 0, & \text{otherwise.} \end{cases}$$

4. Suppose we have jointly continuous X and Y with joint probability density function:

$$f(x, y) = \begin{cases} \frac{4}{y}e^{-4y} & \text{if } 0 < x < y, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) (6 points) Let $f_Y(y)$ denote the probability density function of Y . Find $f_Y(y)$ and identify the distribution of Y by name.

Note. Make sure to identify the parameters of the distribution.

Solution: The marginal PDF is computed as $f_Y(y) = \int_{-\infty}^{\infty} f(x, y)dx$. Note that $f(x, y) = 0$ for $y \leq 0$, so we get $f_Y(y) = 0$ in this case. For $y > 0$ we have

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)dx = \int_0^y \frac{4}{y}e^{-4y}dx = 4e^{-4y}$$

(note that we are integrating a constant). The resulting PDF is exactly the PDF of the $\text{Exp}(4)$ distribution.

- (b) (6 points) Let $f_{X|Y}(x|y)$ denote the conditional density function of X given $Y = y$. Find $f_{X|Y}(x|y)$ and identify the conditional distribution of X given $Y = y$ by name.

Note. Make sure to identify the parameters of the distribution.

Solution: For jointly continuous random variables X, Y the conditional PDF $f_{X|Y}(x|y)$ is defined if $f_Y(y) > 0$, and in that case $f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$. Hence, for $y > 0$ we have

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{\frac{4}{y}e^{-4y}I(0 < x < y)}{4e^{-4y}} = \frac{1}{y}I(0 < x < y).$$

Here $I(0 < x < y)$ is the indicator of $0 < x < y$: it is 1 if $x \in (0, y)$ and 0 otherwise. (One can also write this as a case-defined function.) The resulting conditional PDF is exactly the PDF of the $\text{Uniform}(0, y)$ distribution.

- (c) (3 points) Find $E[X|Y]$.

Solution: We first compute $E[X|Y = y]$. This is well-defined if $f_Y(y) > 0$, which happens if $y > 0$. Since the conditional distribution of X is $\text{Uniform}(0, y)$, and the expectation of the uniform is the midpoint of the interval, we get $E[X|Y = y] = \frac{y}{2}$. From this we get $E[X|Y] = \frac{Y}{2}$.

- (d) (4 points) Find $E[X]$.

Solution: We can compute $E[X]$ by conditioning on Y first and taking the expected value of $E[X|Y]$:

$$E[X] = E[E[X|Y]] = E[Y/2] = \frac{1}{2}E[Y].$$

We have seen that $Y \sim \text{Exp}(4)$, and we know that the expectation of the exponential distribution is one over the parameter, hence

$$E[X] = \frac{1}{2}E[Y] = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}.$$

One can also compute this by evaluating the double integral

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy = \int_0^{\infty} \int_0^y x \frac{4}{y} e^{-4y} dx dy.$$

5. Suppose that X and Y are independent continuous random variables with probability density functions f_X and f_Y .

- (a) (3 points) Write down the integral formula for the probability density function of $Z = X + Y$.

Solution: Because X and Y are independent and continuous, we can use the convolution formula given by

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx.$$

- (b) (7 points) Suppose that $X \sim N(0, 1)$ and $Y \sim \text{Unif}[0, 1]$. Find the probability density function of $Z = X + Y$ at $z = 7$ using the integral formula given in (a).

Note. Your final answer should be in terms of the Φ function.

Solution: By the convolution formula

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(7 - x) dx.$$

We have $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ and $f_Y(y) = I(0 < y < 1)$. Hence $f_X(x) f_Y(7 - x)$ is positive exactly if $0 < 7 - x < 1$, which is equivalent to $6 < x < 7$. In that case the $f_X(x) f_Y(7 - x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, so

$$f_{X+Y}(7) = \int_{-\infty}^{\infty} f_X(x) f_Y(7 - x) dx = \int_6^7 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx.$$

This is the integral of the PDF of the standard normal on $[6, 7]$, which is just the probability of the standard normal being in that interval. We can compute this with the CDF of the standard normal to get $\Phi(7) - \Phi(6)$. Hence

$$f_{X+Y}(7) = \Phi(7) - \Phi(6).$$

6. A monkey is sitting in front of a keyboard that has the 26 capital letters of the alphabet (and nothing else). He starts randomly hitting the keys: he chooses the keys independently and uniformly.

- (a) (4 points) What is the probability that the first two keystrokes are U and W in this order?

Solution: Let A be the event that the first keystroke is U, B the event that the second keystroke is W. By our assumption $P(A) = P(B) = \frac{1}{26}$, and the two events are independent, hence

$$P(\text{first two letters are U and W in this order}) = P(AB) = \frac{1}{26^2}$$

We can also compute this probability by looking at the first two letters, and note that we have 26^2 equally likely outcomes.

- (b) (8 points) We print out the first 1000 keystrokes (in order). Let X be the number of times we see the sequence UW spelled out (these 2 letters next to each other) somewhere in that text. Find $E[X]$.

Solution: Let I_j be the indicator of the event that the j th letter is U and the $(j + 1)$ st letter is W. Then

$$X = I_1 + \cdots + I_{999}.$$

By linearity

$$E[X] = \sum_{j=1}^{999} E[I_j].$$

We have

$$\begin{aligned} E[I_j] &= P(j\text{th letter is U and the } (j + 1)\text{st letter is W}) \\ &= P(1\text{st letter is U and the 2nd letter is W}) \\ &= \frac{1}{26^2} \end{aligned}$$

where we used the fact that the letters are iid, and hence are exchangeable.

This gives $E[X] = \frac{999}{26^2}$.

- (c) (4 points) Find $\text{Var}(X)$ for the random variable X in part (b).

Solution: Using the notation of part (b) and the formula for the variance of the sum of random variables

$$\text{Var}(X) = \text{Var}\left(\sum_{j=1}^{999} I_j\right) = \sum_{j=1}^{999} \text{Var}(I_j) + 2 \sum_{1 \leq i < j \leq 999} \text{Cov}(I_i, I_j).$$

We have $\text{Var}(I_j) = p(1 - p)$ where $p = \frac{1}{26^2}$ is the probability of $I_j = 1$, as computed in part (a).

Note that I_i depends on the i th and $(i + 1)$ st letter, and I_j depends on the j th and $(j + 1)$ st letters. If $i + 1 < j$ then these letters are all different, and hence are independent, which means that in this case $\text{Cov}(I_i, I_j) = 0$.

If $i + 1 = j$ then we need to compute

$$\text{Cov}(I_i, I_{i+1}) = E[I_i I_{i+1}] - E[I_i]E[I_{i+1}].$$

Not that $I_i I_{i+1}$ can only be nonzero if $I_i = I_{i+1} = 1$, which cannot happen. (If $I_i = 1$ then the $(i + 1)$ st letter is W, if $I_{i+1} = 1$ then the $(i + 1)$ st letter is U.) Hence $E[I_i I_{i+1}] = 0$ and

$$\text{Cov}(I_i, I_{i+1}) = E[I_i I_{i+1}] - E[I_i]E[I_{i+1}] = -\frac{1}{26^4}$$

using part (a). Putting everything together

$$\begin{aligned} \text{Var}(X) &= \text{Var}\left(\sum_{j=1}^{999} I_j\right) = \sum_{j=1}^{999} \text{Var}(I_j) + 2 \sum_{1 \leq i < j \leq 999} \text{Cov}(I_i, I_j) \\ &= 999 \text{Var}(I_1) + 2 \sum_{i=1}^{998} \text{Cov}(I_i, I_{i+1}) = 999 \frac{1}{26^2} \left(1 - \frac{1}{26^2}\right) - 2 \cdot 998 \frac{1}{26^4} \end{aligned}$$